

Package #159

apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized thermal spikes that accelerate rain erosion

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'leading-edge erosion' concerning offshore wind turbine blade. User-submitted engineering problem (session input): "Why do leading-edge erosion of offshore wind turbine blades degrade annual energy production, and can a passive surface design extend blade life in rain erosion?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2025-2026). Illustrative records: 'Presentation on the safety considerations with regard to commercial fusion energy'; 'Microgrid implementation can help improve electrical service, reliability, and resilience for localized communities. However, they may be susceptible to damage from natural disasters and extreme weath'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: thermal anomaly response (temperature variation due to surface damage)

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.